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| COMPASS<br>Coastal Observations, Mechanisms, and Predictions Across Systems and Scales | **COMPASS Synoptic Field Sampling Protocol** | |
| Date Created: 4 Apr 2022<br><br>Date Updated: 24 Mar 2025 | Creator Name(s): A. Stearns<br><br>Editor Name(s): A. Stearns | Version: 2 |
| **Tree Plot Installation & Annual Census** | | |

## Objective:

To establish a tree vegetation plot that will allow COMPASS to monitor a representative sample of trees over time. To set forth guidelines for how to conduct the annual census of the trees.

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## I. Experimental Design:

Each COMPASS Synoptic Site will have two (2) tree plots installed, one (1) in the Transition Zone and one (1) in the Upland Zone. Every tree > 5 cm diameter at breast height (DBH) located within the tree plot will be monitored annually for growth and condition status. Shrubs, such as *Morella cerifera* (wax myrtle), that are > 5 cm DBH will also be monitored in the tree

survey. Dead standing trees > 5 cm DBH will be included in the initial Census only. Ideal tree plots will include as many sap flux trees as possible. Each site will have its own sequence of tree tag numbers. There should be no duplicate Tree ID's at any one site.

In 2024 Sweet Hall Marsh added an additional zone, the Swamp. Officially, the originally (2023) designated Upland became Swamp, and the originally (2023) designated Upland Control became Upland where a new tree plot was installed and inventoried.

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## II. Field Safety

### A. Personal Protective Equipment

Close-toed shoes and long pants are required at all times when working at COMPASS sites. Knee-high rubber boots, hip waders, or chest waders are recommended for working in the Transition zones. Chest waders may be preferred for Sweet Hall Marsh due to the condition of the substrate and protection from dense vegetation. Garden or work gloves are recommended for hand safety. Other PPE, such as eye and ear protection, is readily available if desired.

### B. Environmental Awareness

Always conduct a visual site survey prior to, and continuously, during work. This includes:

- Looking up at the canopy, as well as around you at face level, and identifying dead or dying trees, dead trees that may be leaning, and loose or dead limbs that are precariously hanging. These are all hazards to be mentally noted as you are working around or under those areas. If anything appears as a major safety issue, please do not proceed and report back to the Team Lead for further instruction.
- Looking at the ground for potential fallen trees, tripping hazards, etc.
- Looking for anthropogenic hazards and infrastructure such as sampling plots, sensors, cables & wires, etc. that are low to or on the ground. And be mindful when working at or around sapflux trees.
- Assessing the immediate dangers at each tree prior to measurement/recruitment.

- When approaching a tree:
  - Be cautious of poison ivy (a big concern in the SWH and GWI TR zones), smilax, or other thorny vines. Use pruners to snip any obstructions.
  - Assess if the tree is dead or dying. If the tree is dead or dying, be sure that the tree is stable enough to touch and that there is no excessive stem loss at the base that would cause the tree to be unstable during measurement/recruitment

**C.** **COMPASS_CB_FieldSafetyPlan_2024.pdf**

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## III. Materials:

### A. Plot Installation

- (2) 30M+ metric survey tapes
- 5 m D-tape
- Small D-tape
- 1" PVC corner poles - 4/plot per site
- Mallet
- Prunners / vine clippers
- Wax crayons (DBH marker)
- Tree tags (aluminum) - we have designated tags for each plot; make sure you have the correct tags for the site you are installing
- 2" or longer aluminum nails (make sure they fit the tag hole diameter)
- Hammer
- Data collection sheet
- Clipboard
- Pencil
- CB Synoptic Vegetation Field Notebook
- This protocol

### B. Census

- 5 m D-tape

# Tree Plot Installation & Annual Census

- Small D-tape
- DBH Pole
- Prunners / vine clippers
- Wax crayons (DBH marker)
- Tree tags (aluminum) - we have designated tags for each plot; make sure you have the correct tags for the site you are working at
- 2" or longer aluminum nails (make sure they fit the tag hole diameter)
- Hammer
- Garden/work gloves
- Data collection sheet
- Species List / reference materials
- Clipboard
- Pencil
- CB Synoptic Vegetation Field Notebook
- This protocol

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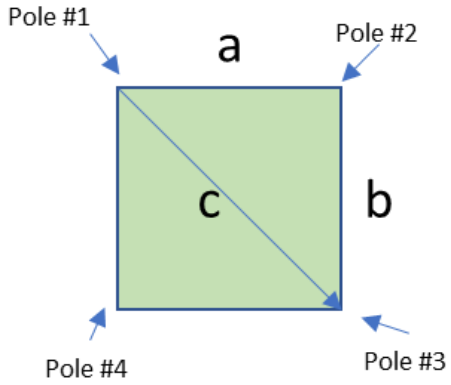
## IV. Procedure:

### A. Plot Installation - Establishing the Plot

Please write notes legibly. If you know your handwriting is questionable, have another team member check your notes for interpretation before leaving site.

1. Designate the general area of the plot.
2. Set corner pole #1
3. Pull tape in the direction you want the distance you want the width
4. Set corner pole #2
5. From corner pole #2, pull tape to the distance desired for the length, approximating a right angle, in the proposed direction of pole #3
6. Use the Pythagorean Theorem to find the hypotenuse

# Tree Plot Installation & Annual Census




$$a^2 + b^2 = c^2$$

$$c = \sqrt{(a^2 + b^2)}$$

7. Pull tape from pole #1 toward the proposed position of pole #3
8. Place pole #3 where the length of the side intercepts the length of the hypotenuse
9. Pull tapes from pole #'s 1 and 3 to find the placement of the last corner, pole #4
10. Make any notes about the installation including the plot dimensions, its general location within the site zone, and any other pertinent information in the Vegetation Field Notebook and installation field data sheet

**B. Initial Census**

*Please write legibly. If you know your handwriting is questionable, have another team member check your notes for interpretation before leaving site.
*Ideally, a two person team is needed for this task. One person to work at the tree and the other person taking the data and serving as an additional visual assessor of environmental conditions and making sure the D-tape is level during measurement.

1. Set up the clipboard with the field collection sheet and fill out the appropriate header information. Write clearly. Write the full names of the team members in case there are questions about this census in the future.
2. Tag and inventory every tree within the plot that is > 5 cm DBH, including dead standing trees.

# Tree Plot Installation & Annual Census

3. Recruit and sequentially tag trees within the plot in a way that makes sense and that will make it easier to navigate during re-census surveys. *E.g. we started tagging in south corner of plot and moved north in 10 m columns.* Be sure to make notes on the field collection sheet about the "path" you took while tagging trees.
    TIP: It is ideal to start from the south and move north since the trees will be tagged on the south side.
4. Approach tree and assess environmental and anthropogenic surroundings, as reviewed in the Field Safety section of this protocol. Don't forget to look up!
5. Identify the height of measure (HOM)
    a. Use the pre-marked DBH pole to locate HOM at DBH, 1.3 m from the ground
    b. If there is an obstruction or deformity on the stem located at or around 1.3 m, find an ideal HOM above or below the obstruction/deformity. *Be sure to note alternate HOM and what the alt HOM is on the field collection sheet*
    c. If the tree is on a slope, the HOM should be taken 1.3 m from the ground surface of the *uphill* side of the tree.
6. Mark the HOM on the tree using a wax crayon
7. Tag the tree on the south side of the tree, opposite side from any sapflux sensors
    a. Nail tag into tree > 10 cm above or below the HOM mark. *This is important since trees have a tendency to swell around nails which would influence the DBH*
    b. Be sure NOT to hammer the nail in too deep in order to leave room for tree growth. This is why a 2" or longer aluminum nail is ideal.
    c. Write the tag number on the field collection sheet
8. Look up to check the tree's status, whether it is in-tact, topped (missing main stem) or missing any major branches.
    a. Document status in the correct column of the field collection sheet along with any other notes about the condition of the tree e.g missing branch, topped
        i. Status Codes:
            LI = Living
            DS = Dead standing
            DC = Dead down
            DN = Dead/not found
    b. If you are unable to determine the status, write "ck status" in the notes column
9. Identify the species and document it in the correct column of the field collection sheet
    a. CB Synoptic Species List

   b. If you are unsure of the species or you cannot ID the species, write “ck sp” in the notes column
   c. For dead trees, you may not be able to ID the species and that is ok. Write “UNKN” in the species column
10. Using a metric diameter tape (D-tape), measure and record the diameter at the designated HOM
   a. For trees < 10 cm DBH use the small DBH tape
   *b.* ***D-tapes have two different measurement sides!! Be sure you are using the diameter side***
   c. Wrap the D-tape around the tree at the previously marked HOM. Peak around to make sure the D-tape is level and is not caught on any bark or branches. The data collector can also serve in this capacity.
   d. Be sure to read the D-tape correctly, especially if it is upside down, which sometimes is the case.
   e. Round measurements down, never up!
   f. Write measurement on field collection sheet
11. Move on to the next tree
12. When finished, conduct a walk-thru of the plot to make sure you did not miss any trees
13. Review field collection sheet:
   a. Is it legible?
   b. Is all header information complete?
   c. Are all of the columns filled in? There should be no blank boxes.
   d. Write any general notes, observations, problems, questions, etc.

**C. Re-census**

*Please write legibly. If you know your handwriting is questionable, have another team member check your notes for interpretation before leaving site.
*Ideally, a two person team is needed for this task. One person to work at the tree and the other person taking the data and serving as an additional visual assessor of environmental conditions and making sure the D-tape is level during measurement.
*Be prepared to tag in new recruits so make sure you have a hammer, nails and the correct tags for the site and zone you are working in.

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1. Set up the clipboard with the field collection sheet and fill out the appropriate header information. Write clearly. Write the full names of the team members in case there are questions about this census in the future.
    a. Check for notes on how the plot was originally tagged and what “path” the installers took
    b. The field collection sheet should relatively go in order of how the plot was tagged
    c. The field collection sheet will have the previous years DBH, status codes, and relevant notes. Review these for each tree you approach
    d. There are also empty lines for new recruits. For new recruits, reference the Initial Census section of this protocol for instruction
2. Approach tree and assess environmental and anthropogenic surroundings, as reviewed in the Field Safety section of this protocol. Don’t forget to look up!
3. Locate tree tag on the south side of the tree and relay to data collector
    a. If you cannot locate a tree, check the previous years status. If it was DS, it likely fell over. Check the ground for fallen trees and check those trees for a tag. You can collect the tag of a fallen, dead tree. Be sure to note that you took the tag!
4. Data collector should confirm species and status at this time and relay any pertinent notes from the previous year(s) such as alt HOM
5. Look up to check the tree's status, whether it is in-tact, topped (missing main stem), missing any major branches, etc.
    a. Document status in the correct column of the field collection sheet along with any other notes about the condition of the tree e.g missing branch, topped
        i. Status Codes:
            LI = Living
            DS = Dead standing
            DC = Dead down
            DN = Dead/not found
    b. If you are unable to determine the status, write “ck status” in the notes column
6. Verify the species
    a. Reference the CB Synoptic Species List
    b. If you are unsure of the species or you cannot ID the species, write “ck sp” in the notes column
7. Locate the HOM wax crayon mark on the south side of the tree
    a. If the mark has faded, re-mark

   b. If you cannot locate the HOM, check notes for alt HOM; if you still cannot locate the HOM mark, use the DBH pole to create a new mark
8. Using a metric diameter tape (D-tape), measure and record the diameter at the designated HOM
   a. For trees < 10 cm DBH use the small DBH tape
   *b.* ***D-tapes have two different measurement sides!! Be sure you are using the diameter side***
   c. Wrap the D-tape around the tree at the previously marked HOM. Peak around to make sure the D-tape is level and is not caught on any bark or branches. The data collector can also serve in this capacity.
   d. Be sure to read the D-tape correctly, especially if it is upside down, which sometimes is the case.
   e. Round measurements down, never up!
   f. Write measurement on field collection sheet
   g. Make sure the DBH is reasonable in relation the previous years DBH
      i. An acceptable reading is within 0.5 cm
      ii. For drastic differences, re-take the measurement
         1. Verify HOM
         2. Check that the tape is not caught on anything and is level
         3. Check you are reading the D-tape properly
         4. Check that the tree is LI (sometimes DBH will decrease if a tree has died)
         5. Check for injury, bark loss, etc
         6. If the measurement is still off, document it and make a note that you "checked" measurement
9. Move on to the next tagged tree
   a. Tag new recruits as you go
10. When finished, conduct a walk-thru of the plot to make sure you did not miss any new recruits
11. Review field collection sheet:
   a. Is it legible?
   b. Is all header information complete?
   c. Are all of the columns filled in? There should be no blank boxes.
   d. Write any general notes, observations, problems, questions, etc.

# Tree Plot Installation & Annual Census

## V. Site Specific Notes:

**A. GWI Tree Plot Install notes** - BBL
Installed by Ben Bond-Lamberty (PNNL) and Lani DuFresne (SERC)
Installation Date: April 6, 2022
Upland plot is 20x20 m
Transition is 15x25 m (long side running parallel to marsh boundary)
Trees are all tagged on the south side at ~1.3 m, and measured ~5 cm above that; measurement point was marked with a wax pencil
Tags start in SW corner of each plot proceed to NW corner; then middle third, again starting on south side, then eastern third
Upland are tagged 617-684; transition are tagged 748-799 and then 142-161

**B. MSM Tree Plot Install Notes**
Installed by Nate McDowell and Jun Yin Ding
Installed on April 11, 2022
No field notes were found for installation. Plot dimensions and HOM will be documented during 2023 Census
Transition Plot is 15x25 m
Upland Plot is 20x20 m
DBH taken at 1.35 m

**C. GCW Tree Plot Install notes** - SW, AES, BBL
Installed by Ben Bond-Lamberty, Alice Stearns, Stephanie Wilson
Installation Date: March 9, 2023
Transition Plot is 25x20 m (long side running parallel to marsh boundary)
Upland Plot will be TEMPEST Control Plot (or a portion of that plot)
Upland Plot is 40x50 m
Trees are tagged on the South-ish side of the tree (facing inland from the waters edge)
Most of the trees were tagged during the PREMIS HSLS Shoreline project. 18 trees had to be tagged (including three SF trees) and some tag numbers are duplicates to existing tagged trees that are OUTSIDE of the GCW_TR Tree Plot so it shouldn't be an issue.

Trees were tagged at 1.4M and measured ~5cm above that
TR tags are 700's and 900's

**D. SWH Tree Plot Install Notes**

Installed by Ben Bond-Lamberty, Stephanie Wilson
Installed on March 29, 2023
Numbering was done by splitting the plot into two 10M wide sections starting at the South end and moving North
Transition Plot is 15x25 m; located East of the boardwalk (behind logger boxes); tags number 601+
Swamp Plot is 20x20 m; tags number 501+ (was Upland 2023)
Tagged at 1.35 m on South side of tree
DBH taken at 1.3 m
Tagged trees >5.0 cm at DBH
Installed by A. Stearns, P. Megonigal
Installed on September 19, 2023
Upland  Plot is 15x15 m; tagged 800+; DBH marked at 1.3 m; no DBH taken 2023 (was Upland Control 2023)

## VI. Resources

**A.** Dominant Tree Species Photos - CB Synoptic Sites
   a. ADD PHOTOS

**B.** Species List - CB Synoptic Sites
   a. COMPASS_CBSynoptic_SpeciesList.pdf

**C.** Tree Plot Inventory - CB Synoptic Sites
   a. CB_TreePlotInventory